

3.4

Consider the z -transform $X(z)$ whose pole-zero plot is as shown in Figure P3.4.

- (a) Determine the ROC of $X(z)$ if it is known that the Fourier transform exists. For this case, determine whether the corresponding sequence $x[n]$ is right sided, left sided, or two sided.
- (b) How many possible two-sided sequences have the pole-zero plot shown in Figure P3.4?
- (c) Is it possible for the pole-zero plot in Figure P3.4 to be associated with a sequence that is both stable and causal? If so, give the appropriate ROC.

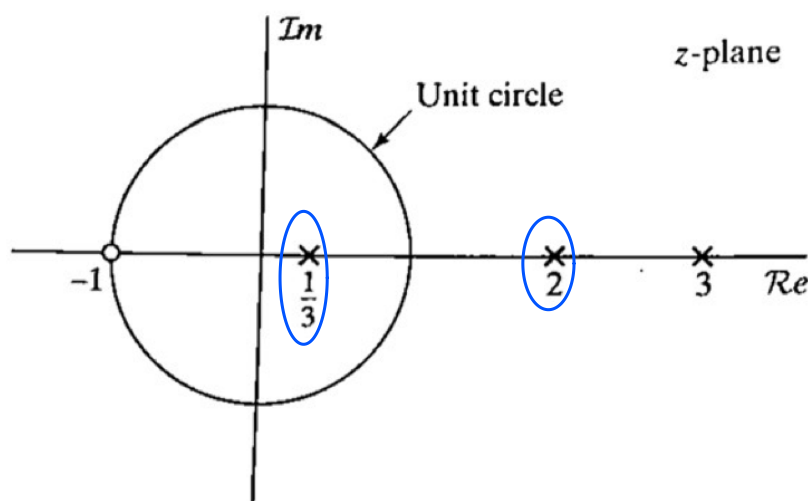


Figure P3.4

- <sol> (a) F.T. exists $\Rightarrow$ ROC: $|\frac{1}{3}| < |z| < |2| \Rightarrow$ two-sided sequence. #
- (b) $|\frac{1}{3}| < |z| < |2|$, $|2| < |z| < |3|$ #
- (c) stable & causal: all poles must be inside unit circle. $\Rightarrow$ no possible ROC. #

3.6

Following are several z -transforms. For each, determine the inverse z -transform using ~~both methods~~—partial fraction expansion and power series expansion—discussed in Section 3.3. In addition, indicate in each case whether the Fourier transform exists.

(a) $X(z) = \frac{1}{1 + \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$

(b) $X(z) = \frac{1}{1 + \frac{1}{2}z^{-1}}, \quad |z| < \frac{1}{2}$

(c) $X(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 + \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}, \quad |z| > \frac{1}{2}$

(d) $X(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{4}z^{-2}}, \quad |z| > \frac{1}{2}$

(e) $X(z) = \frac{1 - az^{-1}}{z^{-1} - a}, \quad |z| > |1/a|$ 長除法

(sol) (a) $a^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-az^{-1}}, |z| > |a|, a = -\frac{1}{2} \Rightarrow x[n] = (-\frac{1}{2})^n u[n], \text{ F.T. exists } \#$

(b) $a = -\frac{1}{2}$, left-sided $\Rightarrow x[n] = -(-\frac{1}{2})^n u[-n-1]$, F.T. doesn't exist.

$$(c) \underline{X}(z) = \frac{1 - \frac{1}{2}z^{-1}}{(1 + \frac{1}{4}z^{-1})(1 + \frac{1}{2}z^{-1})} = \frac{-3}{1 + \frac{1}{4}z^{-1}} + \frac{4}{1 + \frac{1}{2}z^{-1}}, |z| > \frac{1}{2} \Rightarrow \text{right-sided}$$

$\therefore x[n] = (-3) \cdot (-\frac{1}{4})^n u[n] + 4 \cdot (-\frac{1}{2})^n u[n]$, F.T. exists. #

$$(d) \underline{X}(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{4}z^{-2}} = \frac{1}{1 + \frac{1}{2}z^{-1}}, |z| > \frac{1}{2} \Rightarrow \text{right-sided}$$

$\therefore x[n] = (-\frac{1}{2})^n u[n]$, F.T. exists.

$$(e) \underline{X}(z) = -a + \frac{-a^2+1}{-a+z^{-1}} = -a - \frac{a^2(1-a^2)}{1-a^2z^{-1}}, |z| > |a|$$

$\therefore x[n] = -a \cdot \delta[n] - (1-a^2)a^{-(n+1)}u[n]$, F.T. exists #

3.9

A causal LTI system has impulse response $h[n]$, for which the z -transform is

$$H(z) = \frac{1 + z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{4}z^{-1})}$$

(a) What is the ROC of $H(z)$?

(b) Is the system stable? Explain.

(c) Find the z -transform $X(z)$ of an input $x[n]$ that will produce the output

$$y[n] = -\frac{1}{3} \left(-\frac{1}{4}\right)^n u[n] \ominus \frac{4}{3} (2)^n \underline{u[-n-1]}.$$

(d) Find the impulse response $h[n]$ of the system.

<sol> (a) $H(z) = \frac{2}{1 - \frac{1}{2}z^{-1}} + \frac{-1}{1 + \frac{1}{4}z^{-1}}$, $h[n]$ is causal $\Rightarrow \text{ROC: } |z| > \frac{1}{2}$ #

(b) stable: ROC contains unit circle $|z|=1$ #

$$(c) Y(z) = \frac{-\frac{1}{3}}{1 + \frac{1}{4}z^{-1}} + \frac{\frac{4}{3}}{1 - 2z^{-1}}, \frac{1}{4} < |z| < 2$$

$$\Rightarrow \underline{X}(z) = \frac{Y(z)}{H(z)} = \frac{1 - \frac{1}{2}z^{-1}}{1 - 2z^{-1}}, |z| < 2 \quad \frac{1}{1 - 2z^{-1}} + \frac{-\frac{1}{2}z^{-1}}{1 - 2z^{-1}}$$

$$\therefore x[n] = -2^n u[-n-1] + \frac{1}{2} \cdot (2)^{n-1} u[-n] \#$$

$$(d) h[n] = 2 \cdot \left(\frac{1}{2}\right)^n u[n] - \left(-\frac{1}{4}\right)^n u[n] \#$$

3.23

An LTI system is characterized by the system function

$$H(z) = \frac{(1 - \frac{1}{2}z^{-2})}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})}, \quad |z| > \frac{1}{2}.$$

$$\frac{a^{-1}z^{-1}}{1-az^{-1}}$$

$$z^{-1} \downarrow = -a^n u[-n-1] \\ = -a^{n-1} u[-(n-1)-1]$$

(a) Determine the impulse response of the system.

(b) Determine the difference equation relating the system input $x[n]$ and the system output $y[n]$.

$$\text{sol} \rightarrow (a) H(z) = -4 + \frac{5-3z^{-1}}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}} = -4 + \frac{-2}{1-\frac{1}{2}z^{-1}} + \frac{7}{1-\frac{1}{4}z^{-1}}$$

$$\therefore h[n] = -4\delta[n] - 2\left(\frac{1}{2}\right)^n u[n] + 7 \cdot \left(\frac{1}{4}\right)^n u[n] \#$$

$$(b) H(z) = \frac{Y(z)}{X(z)} = \frac{1-\frac{1}{2}z^{-2}}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}} \Rightarrow Y(z)[1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}] = X(z)(1-\frac{1}{2}z^{-2})$$

$$\therefore \text{difference equation: } y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n] - \frac{1}{2}x[n-2] \#$$

3.36

The pole-zero diagram in Figure P3.36 corresponds to the z -transform $X(z)$ of a causal sequence $x[n]$. Sketch the pole-zero diagram of $Y(z)$, where $y[n] = x[-n+3]$. Also, specify the ROC for $Y(z)$.

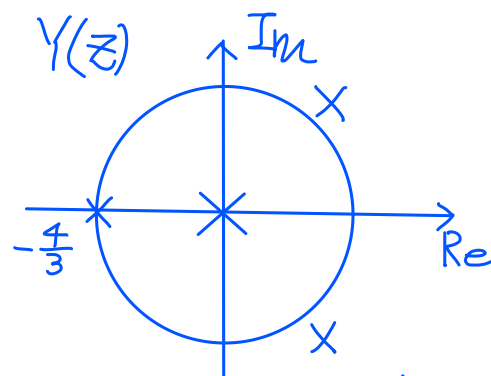
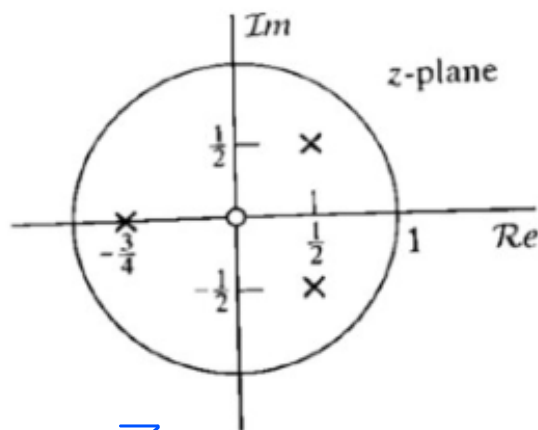


Figure P3.36

$$\text{sol} \rightarrow X(z) = \frac{z}{(z^2 - z + \frac{1}{2})(z + \frac{3}{4})}, \quad |z| > \frac{3}{4}$$

$$\text{而 } y[n] = x[-n+3] = x[-(n-3)] \Rightarrow Y(z) = z^{-3} X(z^{-1})$$

$$Y(z) = \frac{z^{-3} z^{-1}}{(z^{-2} - z^{-1} + \frac{1}{2})(z^{-1} + \frac{3}{4})} = \frac{z^{-4}}{z(2 - 2z + z^2)(\frac{4}{3} + z)} \Rightarrow \text{poles: } 0, -\frac{4}{3}, 1 \pm j$$

Since $x[n]$ is causal, $x[n+3]$ is left-sided, ROC: $0 < |z| < \frac{4}{3}$

Matlab problem

(1) Find the poles and zeros of

$$G(z) = \frac{2z^4 + 16z^3 + 44z^2 + 56z + 32}{3z^4 + 3z^3 - 15z^2 + 18z - 12}$$

(2) Plot the poles and zeros on the z-plane

(3) Find the partial fraction expansion of $G(z)$

(sol>1)

```
>> num=[2 16 44 56 32];
>> den=[3 3 -15 18 -12];
>> [z,p,k]=tf2zp(num,den)
>> disp('zero');disp(z')
zero
-4.0000 + 0.0000i -2.0000 + 0.0000i -1.0000 - 1.0000i -1.0000 + 1.0000i

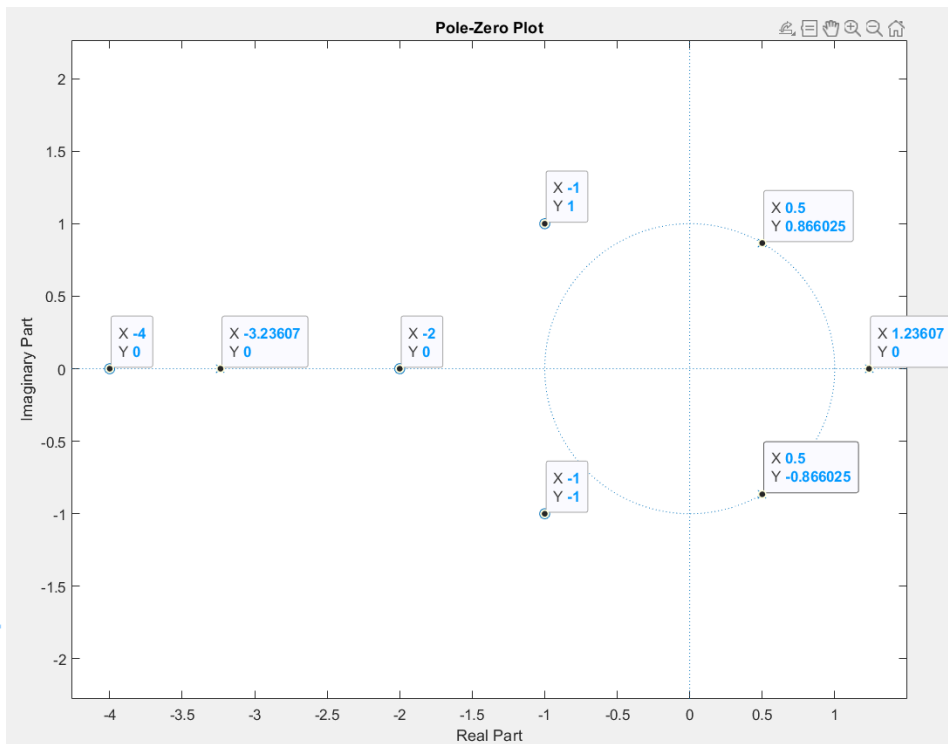
>> disp('pole');disp(p')
pole
-3.2361 + 0.0000i 1.2361 + 0.0000i 0.5000 - 0.8660i 0.5000 + 0.8660i

>> zplane(num,den);
```

```
z =
-4.0000 + 0.0000i
-2.0000 + 0.0000i
-1.0000 + 1.0000i
-1.0000 - 1.0000i

p =
-3.2361 + 0.0000i
1.2361 + 0.0000i
0.5000 + 0.8660i
0.5000 - 0.8660i

k =
0.6667
```



(3)

```
>> [r,p,k]=residuez(num,den);
>> disp('Residues');disp(r')
Residues
-0.0177 + 0.0000i 9.4914 + 0.0000i -3.0702 - 2.3398i -3.0702 + 2.3398i

>> disp('Poles');disp(p')
Poles
-3.2361 + 0.0000i 1.2361 + 0.0000i 0.5000 - 0.8660i 0.5000 + 0.8660i
```

```
>> disp('Constants');disp(k')
```

Constants

-2.6667

$$G(z) = -2.6667 + \frac{-0.0177}{1+3.2361z^{-1}} + \frac{9.4914}{1-1.2361z^{-1}} + \frac{-3.0702-2.3398j}{1-(0.5-0.866j)z^{-1}} + \frac{-3.0702+2.3398j}{1-(0.5+0.866j)z^{-1}}$$